

PseudoRandomness

Pushpendra Pal

June 2025

Reading notes from Justin Thaler's "Proofs Argument and Zero Knowledge"

Keywords:

1 What is PseudoRandomness?

- Perfect secrecy $\xrightarrow{\text{Comprelax}}$ INDISTINGUISHABILITY
- True random $\xrightarrow{\text{Comprelax}}$ Pseudorandomness
- Refers to a distribution of strings. Distribution $\mathcal{D}$ over strings of length l is pseudo random means that $\mathcal{D}$ is IND from uniform distribution over strings of length l .
- And any polynomial time algorithm can **NOT** differentiate between the a string from uniform distribution and string fromm pseudorandom

What is true randomness? Is it relaced to unpredicatability, let's define that. ANS. True randomness is a sequence of bits that cannot be predicted or reproduced, even with complete knowledge of the algorithm used to generate it. It is often derived from physical processes, such as thermal noise or radioactive decay, which are inherently unpredictable.

2 Pseduo Random Function

3 Pseudorandom generator

4 Pseduoradnom permutation